

Retail Analytics Data Platform

End-to-End Data Engineering on Snowflake

Project Report

Submitted as part of the HCLTech Cadre Program

Mentored by

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Project Duration

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Acknowledgement

I would like to express my sincere gratitude to **Mr. Vijender Victor Singh**, my mentor at **HCLTech**, for providing continuous guidance, encouragement, and valuable technical insights throughout my internship under the **HCLTech Cadre Program**. Their mentorship helped me understand industry best practices in data engineering and successfully complete this project.

I would also like to thank the **HCLTech Cadre Program** team for designing a structured learning environment that enabled me to gain practical exposure to enterprise technologies such as Snowflake, SQL, Python, and Streamlit. The hands-on learning experience significantly enhanced my technical and problem-solving skills.

I am grateful for the opportunity to work on the **Retail Analytics Data Platform – End-to-End Data Engineering on Snowflake**, which allowed me to apply theoretical concepts to a real-world, enterprise-oriented project. This experience has strengthened my understanding of modern data engineering practices and prepared me for future professional challenges.

Finally, I extend my heartfelt thanks to everyone who directly or indirectly supported and motivated me during the successful completion of this internship project.

1. Executive Summary

Designed and implemented a **production-grade retail analytics platform** on Snowflake, covering the full data lifecycle from raw ingestion through governed, analytics-ready datasets and interactive dashboards. The project demonstrates enterprise data engineering best practices across 13 modular SQL scripts and a 5-page Streamlit application.

13

SQL Modules

3

Data Layers

9

Source Tables

5

Dashboard Pages

3

Fact Tables

4

Dimension Tables

Business Impact: This platform enables real-time retail decision-making — from store-level sales optimization and inventory replenishment to customer lifetime value analysis and executive KPI monitoring.

2. Architecture Overview

Medallion Architecture (Bronze → Silver → Gold)

BRONZE: Raw Ingestion

SILVER: Cleaned & Typed

GOLD: Star Schema

STREAMLIT: **BI Dashboard**

Infrastructure Components

Component	Purpose	Configuration
RETAIL_INGEST_WH	Data loading (COPY INTO)	X-Small, Auto-suspend 60s
RETAIL_TRANSFORM_WH	Silver/Gold transformations	X-Small, Auto-suspend 60s
RETAIL_BI_WH	Dashboard queries	X-Small, Multi-cluster (1-2), Economy
RETAIL_DB	Central database	5 schemas (Bronze, Silver, Gold, Governance, Monitoring)

Role-Based Access Control (RBAC)

Role	Responsibility	Access Level
RETAIL_ADMIN	Platform owner	Full DB + all schemas
RETAIL_ENGINEER	Data engineering / ETL	Create objects in all schemas
RETAIL_ANALYST	Analytics / reporting	SELECT on Gold layer
RETAIL_BI	BI tool service accounts	SELECT on Gold views (row-filtered)

3. Data Pipeline — Module Breakdown

The pipeline is organized into 14 sequentially numbered SQL files, each handling a distinct responsibility:

#	Module
00	Project Setup Roles, warehouses, database, schemas, grants
01	File Formats & Stage CSV format, internal stage with directory table
02	Bronze DDL 9 raw tables + load rejects table (all STRING for flexibility)
03	COPY INTO Bronze Bulk load from stage with error handling
04	Silver DDL 9 typed, constrained tables with PK/audit columns

#	Module
05	Silver Transform MERGE (SCD Type 1), data cleansing, hash-based change detection
06	Gold Star Schema 4 dimensions (incl. SCD2 customer) + 3 fact tables
07	Incremental Streams & Tasks CDC with append-only streams + scheduled stored procedures
08	Governance & Security Masking policies, row-access policy, data classification tags
09	BI Views & KPIs 4 secure/analytical views for dashboard consumption
10	Data Quality Tests PK uniqueness, FK integrity, reconciliation, constraints
11	Performance & Cost Monitoring Resource monitors, expensive query audit, credit tracking
12	Time Travel, Clone, Share, Unload Zero-copy clone, time travel, secure sharing, COPY INTO export
13	Dynamic Tables Declarative pipeline alternative to streams/tasks

4. Data Model — Star Schema

Fact Tables

Table	Grain	Measures
FACT_SALES	One row per order line item	quantity, gross_sales, discount, net_sales, tax, total
FACT_RETURNS	One row per return	return_amount, return_reason
FACT_INVENTORY_DAILY	One row per store-product-day	on_hand_qty, reorder_level, low_stock_flag

Dimension Tables

Table	Type	Key Attributes
DIM_DATE	Generated (2024-2027)	year, quarter, month, week, day, is_weekend
DIM_STORE	Type 1	store_name, city, state, region
DIM_PRODUCT	Type 1	product_name, category, brand, list_price
DIM_CUSTOMER	Type 2 (SCD2)	name, email, loyalty_tier, effective_from/to, is_current
DIM_PROMOTION	Type 1	promotion_code, discount_type, discount_value

5. Data Governance & Security

Control	Implementation	Scope
Email Masking	Dynamic masking policy — shows full email to ADMIN/ANALYST, masked to others	DIM_CUSTOMER.EMAIL
Phone Masking	Shows last 4 digits only (except ADMIN)	DIM_CUSTOMER.PHONE
Row-Level Security	Region-based row access policy — RETAIL_BI sees only "South" region	DIM_STORE
Data Classification	PII tags on email and phone columns	DIM_CUSTOMER
Resource Monitor	50-credit monthly cap; notify at 75%, suspend at 90%	All warehouses

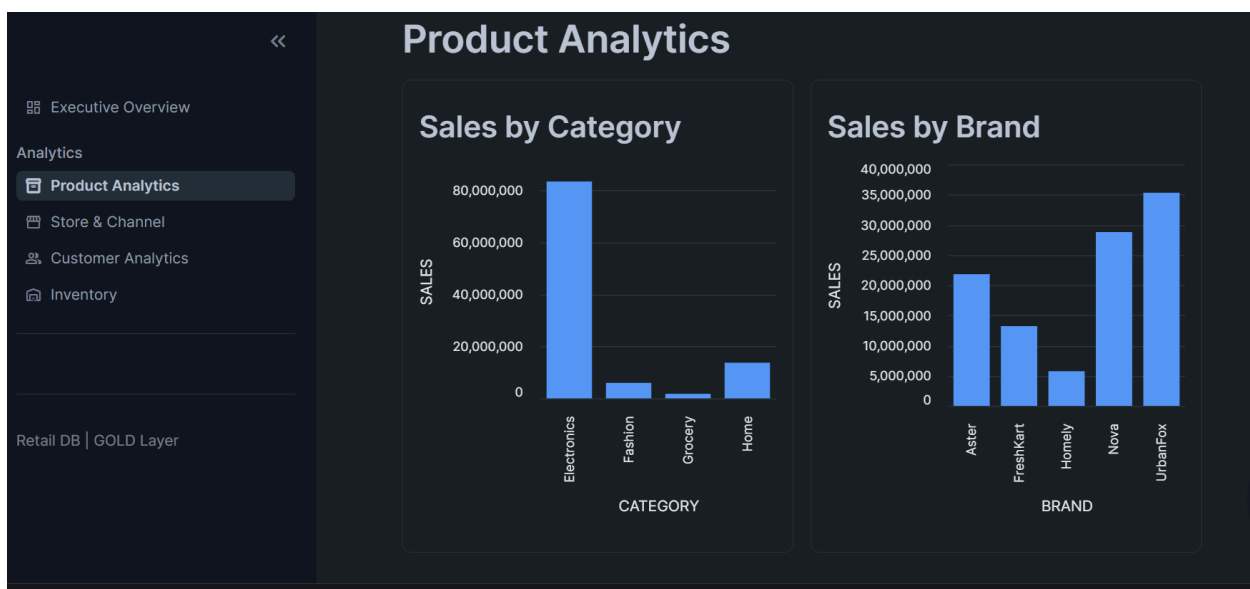
6. Streamlit Analytics Dashboard

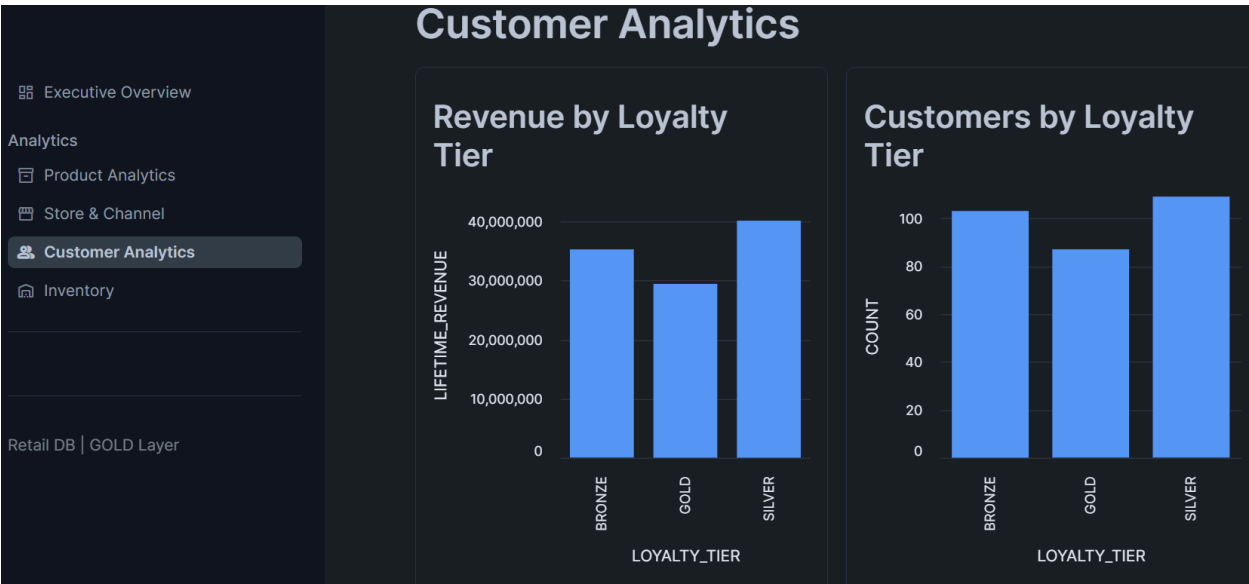
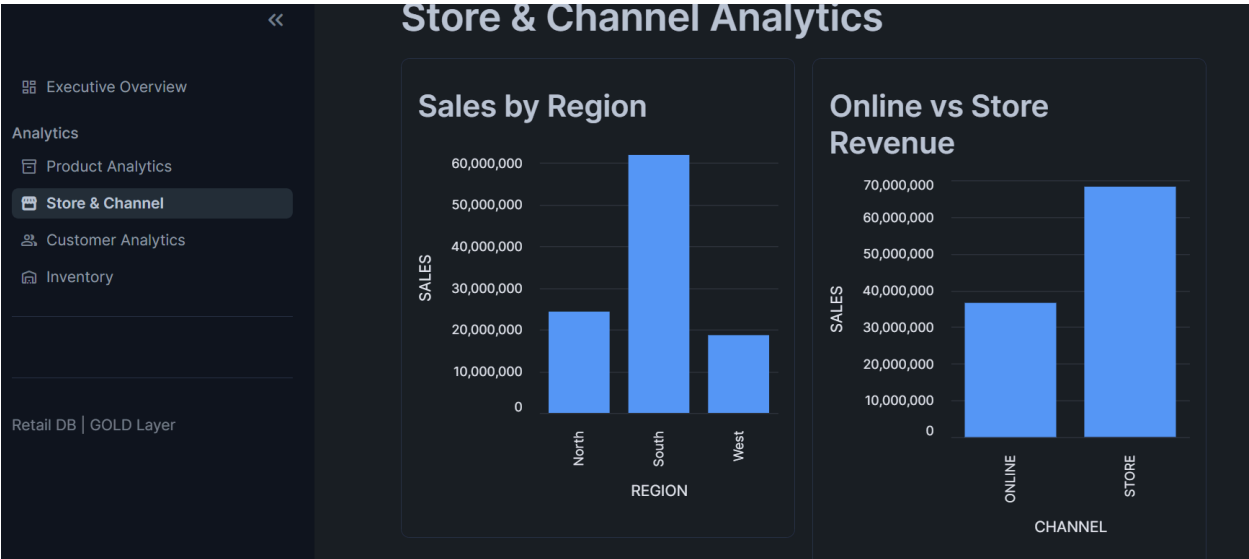
A **5-page interactive Streamlit application** deployed on Snowflake, querying the Gold layer directly:

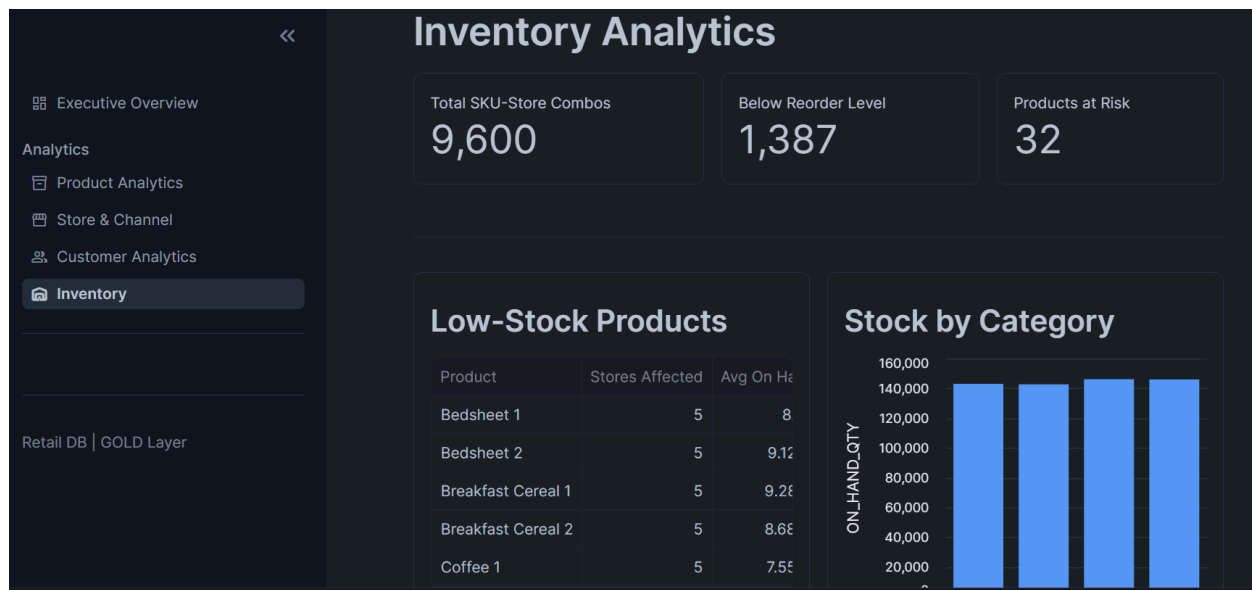
Page	Key Visuals & Metrics
Executive Overview	Net Sales, Orders, Units, AOV, Discount Rate, Returns, MoM Growth, Daily Sales Trend with date filter
Product Analytics	Sales by Category/Brand, Top/Bottom 10 Products, Return Rate, Discount Impact
Store & Channel	Revenue by Region, Online vs Store, Store Ranking, AOV by Channel, Contribution %

Page	Key Visuals & Metrics
Customer Analytics	Revenue by Loyalty Tier, Repeat Rate, Top Customers (LTV), Geographic Distribution
Inventory	Low-Stock Alerts, Stock by Category/Store, Inventory Concentration, Lost-Sales Indicators

Technical Highlights: Multi-page navigation with sidebar. Responsive layout with horizontal containers, and real-time query against live Snowflake tables.







7. Advanced Snowflake Features Demonstrated

Change Data Capture (CDC)

Append-only streams on Bronze tables detect new records; scheduled tasks trigger stored procedures for incremental MERGE into Silver.

Dynamic Tables

Declarative alternative to streams/tasks with `TARGET_LAG='15 minutes'` — Snowflake auto-refreshes the materialization.

Zero-Copy Cloning

Instant dev/test environment via `CREATE DATABASE ... CLONE` with no additional storage cost.

Secure Data Sharing

Created a share with secure views for cross-account analytics distribution without data movement.

Time Travel

Point-in-time queries and table restoration for data recovery scenarios.

Data Quality Framework

Automated tests for PK uniqueness, FK integrity, positivity constraints, and cross-layer reconciliation.

8. Skills & Competencies Demonstrated

Domain	Skills
Data Engineering	ETL/ELT pipeline design, Medallion architecture, incremental loading, MERGE operations, hash-based change detection, data type casting & validation
Data Modeling	Dimensional modeling (Kimball), star schema, SCD Type 1 & 2, surrogate keys, date dimension generation, fact table grain definition
Snowflake Platform	Streams, Tasks, Dynamic Tables, Stages, File Formats, COPY INTO, Time Travel, Zero-Copy Clone, Secure Sharing, Resource Monitors
Data Governance	Dynamic data masking, row access policies, data classification tags, RBAC with least-privilege roles
Data Quality	Automated DQ tests, reconciliation checks, referential integrity validation, monitoring tables
Cost Optimization	Multi-cluster warehouses, auto-suspend, resource monitors with tiered alerts, query performance analysis
BI / Analytics	KPI design, secure views, Streamlit dashboarding, multi-page app architecture, interactive filtering
Python	Streamlit, Pandas, data manipulation, Snowflake connector

9. Key Learnings & Takeaways

What I Learned

- **End-to-end ownership:** Designed the complete data lifecycle — from raw file landing to executive-facing dashboards — reinforcing the importance of thinking holistically about data platforms.
- **Separation of concerns:** The Medallion pattern (Bronze/Silver/Gold) keeps raw data immutable, transformations auditable, and analytics performant.
- **Governance from Day 1:** Implementing masking policies and RBAC early is far easier than retrofitting them later. Security is an architecture decision, not an afterthought.
- **Incremental vs. batch trade-offs:** Streams + Tasks provide event-driven freshness while Dynamic Tables offer simplicity — knowing when to use each is critical.

- **Cost awareness:** Proper warehouse sizing, auto-suspend, and resource monitors prevent runaway costs — essential for production systems.
- **Data quality as code:** Automated DQ tests catch issues before they reach dashboards, building trust in the data.
- **Zero-copy cloning:** Enables safe experimentation without duplicating storage — a game-changer for dev/test workflows.

10. Conclusion

This project represents a **complete, production-grade data platform** built from scratch on Snowflake. It demonstrates not just technical SQL and Python skills, but a mature understanding of:

- How enterprise data platforms are architected and maintained
- How to balance performance, cost, and governance
- How to deliver actionable business insights through self-service analytics
- How to build systems that are auditable, recoverable, and scalable

The platform is ready to scale: adding new data sources means extending Bronze, new transformations flow naturally to Silver, and the Gold star schema can accommodate additional fact/dimension tables without restructuring the entire pipeline.

(Built with Snowflake, Streamlit, SQL and Python)

RIJUL SINGHAL